

SMEDEREVSKA PALANKA, Serbia

WMO: 132790

Lat: **44.367N**Lon: **20.950E**Elev: **122**StdP: **99.87**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.0	-8.0	-13.5	1.2	-10.2	-10.6	1.5	-7.2	8.1	1.8	7.3	2.4	1.6	320	0.441

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	34.4	21.1	32.6	21.0	30.8	20.5	22.8	30.8	22.0	29.8	21.2	28.8	2.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.5	15.4	26.1	19.5	14.5	25.0	18.7	13.7	24.1	68.0	30.8	64.9	30.0	62.0	28.8	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.3	6.2	5.3	-16.4	37.3	4.9	2.5	-19.9	39.1	-22.8	40.6	-25.5	42.0	-29.0	43.8		
(5)			-16.7	24.2	4.7	1.0	-20.1	25.0	-22.8	25.6	-25.4	26.1	-28.8	26.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.2	1.1	3.1	7.3	12.4	16.9	20.5	22.4	22.4	17.6	12.3	7.6	2.3
	DBStd	8.64	5.10	5.33	4.71	4.00	3.59	3.51	3.11	3.12	3.79	4.41	4.76	4.80
	HDD10.0	978	278	198	109	22	1	0	0	0	0	28	100	242
	HDD18.3	2651	536	427	342	181	71	18	4	5	58	191	321	497
	CDD10.0	1783	1	5	25	93	215	314	385	384	228	99	29	3
	CDD18.3	413	0	0	0	2	26	83	131	131	36	4	0	0
	CDH23.3	4662	0	0	2	47	313	926	1483	1459	389	44	0	0
	CDH26.7	1862	0	0	0	7	65	342	653	664	127	4	0	0
Wind	WSAvg	2.2	2.3	2.4	2.6	2.5	2.1	2.0	2.0	1.9	2.0	2.0	2.3	2.3
Precipitation	PrecAvg	664	43	41	46	54	65	75	67	54	60	57	47	53
	PrecMax	1035	86	77	116	129	238	152	249	132	158	118	114	129
	PrecMin	379	10	9	4	2	19	24	8	0	12	1	4	7
	PrecStd	140	21	21	31	27	44	33	50	36	35	34	24	30
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.6	19.0	22.7	27.5	30.5	34.2	37.0	37.3	33.6	27.0	21.4	15.5
		MCWB	9.9	11.2	12.8	16.7	18.9	21.7	21.1	21.4	20.1	17.5	14.6	11.0
	2%	DB	11.9	15.9	20.2	24.3	28.4	32.2	34.3	34.8	30.1	24.4	19.1	13.2
		MCWB	8.3	9.9	11.7	14.8	18.5	21.6	21.3	20.9	19.0	16.7	13.4	9.7
	5%	DB	10.0	13.2	17.7	22.1	26.8	30.4	32.4	32.5	27.5	22.1	16.9	11.1
		MCWB	7.3	8.6	10.9	13.8	17.9	20.9	21.1	20.7	18.2	15.8	12.2	8.2
	10%	DB	8.1	10.9	15.3	20.1	25.0	28.6	30.4	30.5	25.2	19.9	14.8	9.1
		MCWB	6.0	7.4	9.6	12.9	17.2	20.1	20.7	20.3	17.4	14.7	11.1	6.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.1	11.8	13.8	17.4	20.3	24.3	23.6	24.2	21.2	18.9	15.2	11.8
		MCDB	14.3	17.6	21.2	25.9	27.8	31.1	31.0	31.4	30.0	25.4	20.4	14.4
	2%	WB	8.6	10.2	12.4	15.7	19.2	22.6	22.7	22.7	19.7	17.3	13.7	9.9
		MCDB	11.4	15.2	18.8	22.9	26.7	29.9	30.7	30.7	27.6	22.8	18.4	12.8
	5%	WB	7.4	9.0	11.2	14.5	18.5	21.6	21.9	21.7	18.8	16.1	12.6	8.4
		MCDB	9.7	12.8	17.0	20.7	25.5	28.6	29.8	29.8	25.9	21.0	16.3	10.9
	10%	WB	6.1	7.7	10.0	13.4	17.5	20.6	21.1	20.9	18.0	15.1	11.3	7.0
		MCDB	8.1	10.8	14.8	18.8	23.9	27.4	28.7	28.7	24.0	19.4	14.5	8.8
Mean Daily Temperature Range	5% DB	MDBR	6.6	8.1	10.0	11.0	11.5	11.6	12.6	13.1	11.4	10.5	8.2	6.2
		MCDBR	9.7	12.8	14.7	15.4	15.1	15.3	16.6	17.6	15.6	14.0	11.4	9.4
		MCWBR	6.7	7.8	7.8	7.6	6.5	6.0	5.6	5.8	6.2	7.2	6.6	6.4
	5% WB	MCDBR	9.2	11.8	13.1	13.3	13.6	13.6	13.6	14.0	13.4	12.8	10.3	8.8
		MCWBR	6.5	7.3	7.2	6.8	6.3	5.7	5.2	5.3	5.7	6.9	6.0	6.2
Clear-Sky Solar Irradiance	taub	0.344	0.371	0.405	0.463	0.451	0.445	0.448	0.460	0.436	0.398	0.360	0.341	
	taud	2.322	2.225	2.198	2.053	2.148	2.207	2.200	2.162	2.197	2.298	2.354	2.347	
	Ebn at Noon	756	798	821	800	824	829	822	795	779	760	731	719	
	Edh at Noon	82	108	126	158	148	141	140	140	125	100	80	74	
All-Sky Solar Radiation	RadAvg	1.27	2.09	3.29	4.49	5.45	6.09	6.20	5.52	3.87	2.65	1.63	1.09	
	RadStd	0.18	0.27	0.41	0.48	0.56	0.60	0.48	0.53	0.41	0.32	0.16	0.16	